

Create an Azure Cosmos DB for NoSQL Account

Summary:-

In this lab, you will create an Azure Cosmos DB account using the Azure Cosmos DB for NoSQL API. You will configure the account with the required workload type, resource group, account name, region, and capacity mode, enable the free tier, configure networking, and deploy the Cosmos DB account using the Azure portal.

Let's Start:-

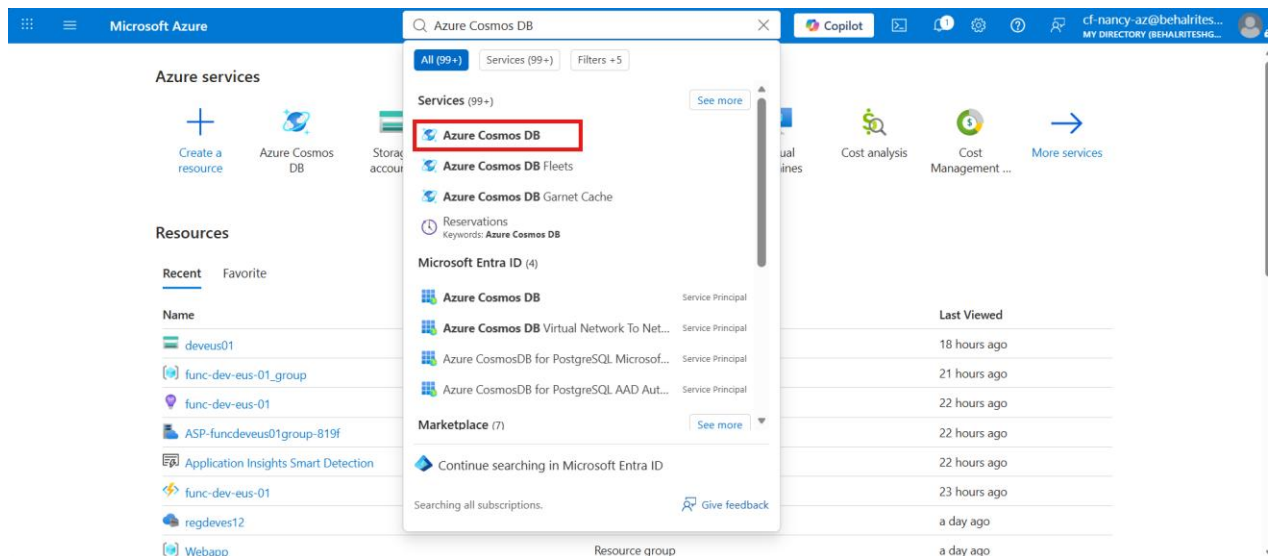
After completing this lab, you will be able to:

- Create an Azure Cosmos DB account.
- Select the Azure Cosmos DB for NoSQL API.
- Configure the account settings and capacity mode.
- Enable the free tier for Azure Cosmos DB.
- Deploy and verify the Cosmos DB account.

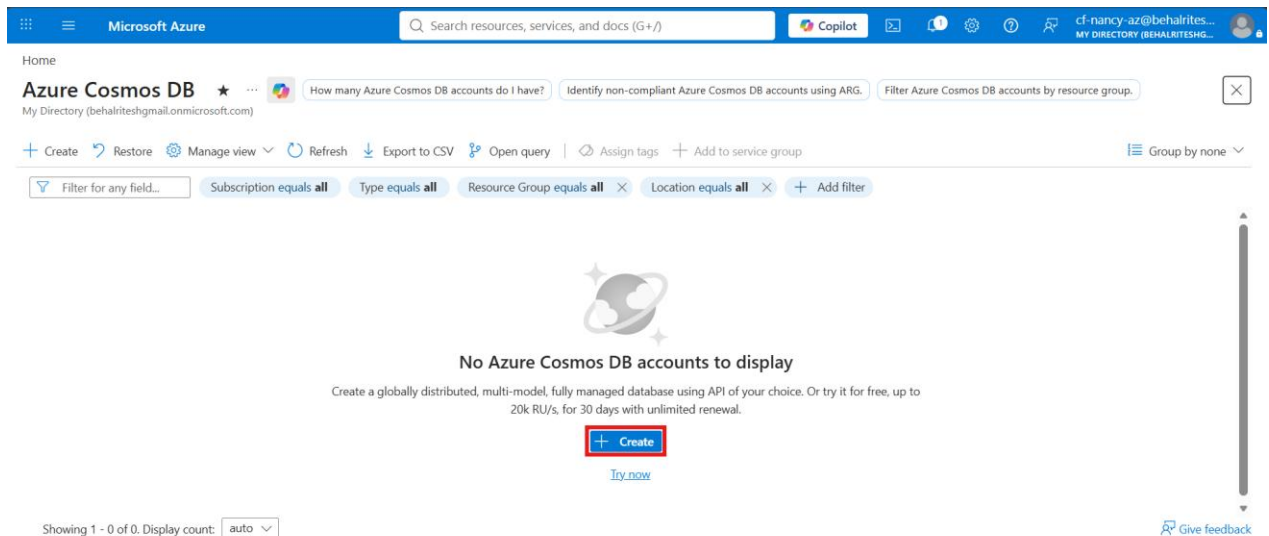
Lab Steps:-

Step 1 – Open Azure Cosmos DB for NoSQL

- Sign in to the Azure Portal.
- Search for **Azure Cosmos DB**.



- Select **Azure Cosmos DB**.



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Home

Azure Cosmos DB

My Directory (behalritesghmail.onmicrosoft.com)

How many Azure Cosmos DB accounts do I have? Identify non-compliant Azure Cosmos DB accounts using ARG. Filter Azure Cosmos DB accounts by resource group.

Create Restore Manage view Refresh Export to CSV Open query Assign tags Add to service group Group by none

Filter for any field... Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

No Azure Cosmos DB accounts to display

Create a globally distributed, multi-model, fully managed database using API of your choice. Or try it for free, up to 20k RU/s, for 30 days with unlimited renewal.

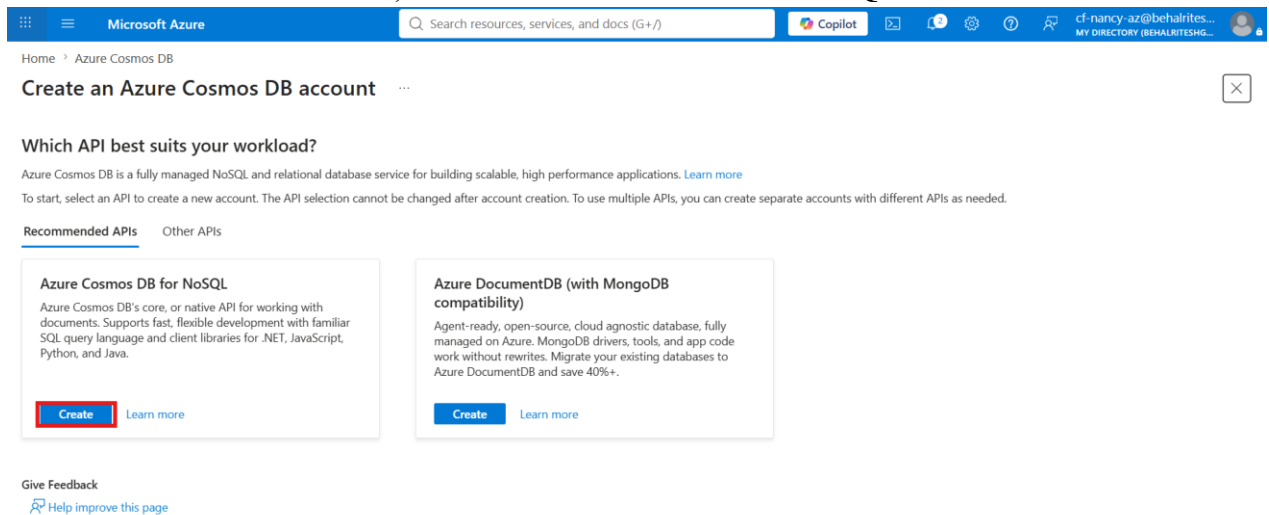
Create

Try now

Showing 1 - 0 of 0. Display count: auto

Give feedback

- Click **Create**.
- Under **Recommended**, select **Azure Cosmos DB for NoSQL**.



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Create an Azure Cosmos DB account

Which API best suits your workload?

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. [Learn more](#)

To start, select an API to create a new account. The API selection cannot be changed after account creation. To use multiple APIs, you can create separate accounts with different APIs as needed.

Recommended APIs Other APIs

Azure Cosmos DB for NoSQL

Azure Cosmos DB's core, or native API for working with documents. Supports fast, flexible development with familiar SQL query language and client libraries for .NET, JavaScript, Python, and Java.

Create Learn more

Azure DocumentDB (with MongoDB compatibility)

Agent-ready, open-source, cloud agnostic database, fully managed on Azure. MongoDB drivers, tools, and app code work without rewrites. Migrate your existing databases to Azure DocumentDB and save 40%+.

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- Click **Create**.
- This step creates a Cosmos DB account that uses the NoSQL API.

Step 2 – Configure the Basics

- Configure the following settings:
 - **Workload Type:** Learning
 - **Subscription:** Select your Azure subscription.
 - **Resource Group:** Select your existing resource group or create a new.

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Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Basics Global distribution Networking Backup Policy Security Tags Review + create

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. Go to production starting at \$24/month per database, multiple containers included. [Learn more](#)

Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type *

Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource Group *

[Create new](#)

Instance Details

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- **Account Name:** Enter a unique name.
- **Region:** Select East US/ East US 2.

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Instance Details

Account Name *

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Availability Zones ☐ Enable ☒ **Disable**

Location *

Available locations are determined by your subscription's access and availability zone support (if that is enabled). If you don't see or cannot select your desired location, please open a support request for region access.
[Click here for more details on how to create a region access request](#)

Capacity mode ☒ **Provisioned throughput** ☐ Serverless

Request units per second (RU/s) are pre-configured and billed hourly, offering guaranteed throughput with two modes—**Autoscale**, which dynamically adjusts within a 1-10x range for varying workloads, and **Manual**, with fixed RU/s for steady, predictable traffic.

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- Under **Capacity Mode**, select **Provisioned Throughput**.
- Enable **Apply Free Tier Discount**.

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[Click here for more details on how to create a region access request](#)

Info Unsure of your traffic patterns? Explore Azure Cosmos DB's serverless offering if you're just starting out and skip capacity planning by only paying for the request units (RUs) you consume. [Learn more](#)

Capacity mode

☒ **Provisioned throughput**
Request units per second (RU/s) are pre-configured and billed hourly, offering guaranteed throughput with two modes— **Autoscale**, which dynamically adjusts within a 1-10x range for varying workloads, and **Manual**, with fixed RU/s for steady, predictable traffic.

☐ **Serverless**
Ideal for intermittent or unpredictable traffic. Billed only for consumed RU/s, scaling on-demand without capacity planning or minimum charges.
[Learn more about capacity mode](#)

With Azure Cosmos DB free tier, you will get the first 1000 RU/s and 25 GB of storage for free in an account. You can enable free tier on up to one account per subscription. Estimated \$64/month discount per account.

Apply Free Tier Discount ☒ **Apply** ☐ Do Not Apply

Limit total account throughput ☒ Limit the total amount of throughput that can be provisioned on this account
Info This limit will prevent unexpected charges related to provisioned throughput. You can update or remove this limit after your account is created.

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- Click **Next**.
- This step configures the basic settings for the Cosmos DB account.

Step 3 – Configure Global Distribution and Networking

- On the **Global Distribution** tab, leave all settings at their default values.

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Basics **Global distribution** Networking Backup Policy Security Tags Review + create

Configure global distribution and regional settings for your account. You can also change these settings after the account is created.

Geo-Redundancy ☐ Enable ☒ **Disable**

Multi-region Writes ☐ Enable ☒ **Disable**

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- Click **Next**.
- On the **Networking** tab:
 - Select **Enable access from all networks**.

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Basics Global distribution **Networking** Backup Policy Security Tags Review + create

Network connectivity

You can connect to your Azure Cosmos DB account either publically, via public IP addresses or service endpoints, or privately, using a private endpoint.

Connectivity method *

- ☒ All networks
- ☐ Public endpoint (selected networks)
- ☐ Private endpoint

All networks will be able to access this CosmosDB account. <http://aka.ms/network-security>

Connection Security Settings

Minimum Transport Layer Security Protocol TLS 1.2

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- Click **Next**.
- This step configures the networking settings for the Cosmos DB account.

Step 4 – Configure Backup and Security

- On the **Backup Policy** tab, leave the default settings.

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Basics Global distribution Networking **Backup Policy** Security Tags Review + create

Azure Cosmos DB provides three different backup policies. You will not be able to switch to Periodic mode once you adopt Continuous mode. [Learn more](#) about the differences of the backup policies and pricing details.

Backup policy

- ☒ Periodic
Backup is taken at periodic interval based on your configuration
- ☐ Continuous (7 days)
Provides backup window of 7 days / 168 hours and you can restore to any point of time within the window. This mode is available for free.
- ☐ Continuous (30 days)
Provides backup window of 30 days / 720 hours and you can restore to any point of time within the window. This mode has cost impact.
- ☐ Continuous (35 days)
Provides backup window of 35 days / 840 hours and you can restore to any point of time within the window. This mode has cost impact.

Backup interval 240 Minute(s)
60-1440

Backup retention 8 Hours(s)
8-720

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- Click **Next**.
- On the **Security** tab, leave all settings at their default values.

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Basics Global distribution Networking Backup Policy **Security** Tags Review + create

Key-based Authentication

Key-based authentication uses account keys that are granted permissions to the entire Cosmos DB account. Disabling these methods improves security by ensuring that Cosmos DB database accounts exclusively require Entra ID identities for authentication. [Learn More](#)

Key-based Authentication ☒ Enable ☐ Disable

Data Encryption

Azure Cosmos DB encryption protects your data at rest by seamlessly encrypting your data as it's written in our datacenters, and automatically decrypting it for you as you access it.

By default your Azure Cosmos DB account is encrypted at rest using service-managed keys. At the moment, you will not be able to switch back to service-managed key after opting into using custom-managed key while creating your account. [Learn More](#)

Data Encryption * ☒ Service-managed key ☐ Customer-managed key (CMK)

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- Click **Next**.
- On the **Tags** tab also, leave all settings at their default values.

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Basics Global distribution Networking Backup Policy Security **Tags** Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [learn more](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Key	Value	Resource Type
		Azure Cosmos DB account

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- Click **Review + Create**.
- This step applies the default backup and security configuration.

Step 5 – Create the Cosmos DB Account

- Review the configuration.
- Click **Create**.

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Validation Success

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Creation Time

Estimated Account Creation Time (in minutes) 3

The estimated creation time is calculated based on the location you have selected

Basics

Subscription	Visual Studio Enterprise Subscription
Resource Group	(new) rg-az204-dev-eus
Location	East US 2
Account Name	(new) app-cosmos-db-01-acc
API	Azure Cosmos DB for NoSQL
Capacity mode	Provisioned throughput

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- Wait for the deployment to complete successfully.
- This step deploys the Azure Cosmos DB for NoSQL account.

Verification

Verify the following:

- The Azure Cosmos DB account is created successfully.
- The account uses the **Azure Cosmos DB for NoSQL** API.
- The configured resource group and region are correct.
- The **Provisioned Throughput** capacity mode with **Free Tier** is enabled.
- The deployment completes successfully and the Cosmos DB account is available in the Azure portal.